Besides the actual values, notice some patterns repeating…

The point is that we wouldn't be able to that for larger instances: with the full graph in figure 17.8, for example, it already takes several minutes to generate all 3.6 million possible permutations!

17.2.3 Pruning the Domain

The reason we wanted to show you the cost function on the full domain of the sub-problem with 6 vertices is because this chart can teach us a lot. For instance, we can see there are many local minima that appears to have the same cost: can you guess why is that?

As always, take a couple of minutes (if you'd like) to think about the answer, before moving on to the solution.

To answer the question, consider these permutations of the cities in the graph:

```
[Opal City, Civic City, New Carthage, Happy Harbor,
Gotham City, Metropolis]
```

```
[New Carthage, Happy Harbor, Gotham City,
Metropolis, Opal City, Civic City]
```

What's difference between these two permutations, in terms of solutions? Since we are considering closed tours (from the first city, through all other cities, and then back to the start), they are completely identical, except the second one is